ABDULLAH ALAFARI

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EDUCATION & TRAINING

Prince Sultan University Expected 2028
Bachelor of Science in Computer Science Major GPA: 3.81/4.00 Cumulative GPA: 3.51/4.00

Tuwaiq Academy – Engineering Solutions Design & Manufacturing, Advanced Camp Jun 2026 – Sep 2026
Tuwaiq Academy – Mechatronics Bootcamp Jan 2026 – Feb 2026

WORK EXPERIENCE

Intern Jun 2025 – Jul 2025
Mawhiba – King Abdulaziz and His Companions Foundation for Giftedness and Creativity *Riyadh*

· Tested the Mawhiba Alumni website and admin panel for bugs and usability issues; around 20 were fixed

PROJECTS

Khzin – Smart Water Tank Monitor Jun 2026 – Sep 2026
ESP32-C6, Zigbee, I2C, KiCad, Home Assistant

· Built a water level sensor on an ESP32-C6 with a pressure sensor read over I2C, calibrated on a real tank
· Built Zigbee communication between the sensor and a base unit, with remote settings and OTA updates
· Designed the PCB in KiCad and a 3D-printed enclosure; added a smart home dashboard and Telegram alerts
· Deployed on a household tank, running continuously for one month

Home Lab – Self-Hosted Smart Home & Network 2023 – Present
Home Assistant, Zigbee2MQTT, MQTT, Docker, Synology NAS, WireGuard, Tailscale

· Run a fully local smart home across a 3-floor house with 82 Zigbee devices, used daily by the family
· Migrated the whole house off cloud-dependent hubs to local control over Zigbee and MQTT
· Segmented the network into IoT, guest, and household VLANs, with remote access over VPN
· Self-host Docker services on a Synology NAS, including Pi-hole for network-wide DNS filtering

Car Steering Wheel as a Nintendo Switch Controller Oct 2025
Raspberry Pi, Python, ADB, Bluetooth (nxbt)

· Parsed live steering angle and button events from a Haval Jolion's Android head unit via ADB logcat
· Emulated a Switch controller over Bluetooth with nxbt, turning the car's wheel into a game controller

Minecraft–Smart Home Bridge (Open Source, Python) Apr 2026

· Python tool, installable with pip, that lets a button pressed inside Minecraft control real smart home devices

LEADERSHIP & AWARDS

President – PSU Game Development Club 2025 – 2026

· Organized workshops on Unity and game-jam preparation

PSU Capture the Flag (CTF) Competition – 1st Place out of 10 teams Nov 2025

SKILLS

Languages: Python, C++, Java
Embedded: Microcontroller firmware, embedded Linux, I2C, UART, PWM
Networking & IoT: Zigbee, MQTT, TCP/IP, VLANs, VPNs
Systems & Tools: Linux, Docker, Git (version control)
Hardware & Design: Oscilloscope, SMD soldering, circuit prototyping, 3D design and printing